

ENDURANCE TIER • SOLAR • BATTERY • HVO GENERATOR

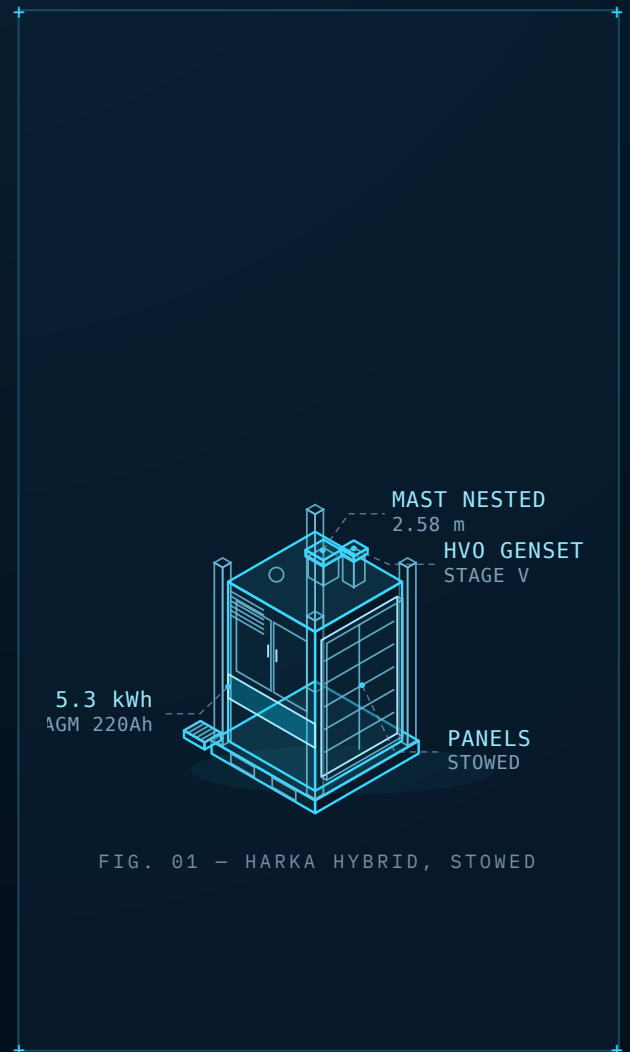
Harka Hybrid

An autonomous 6 m mast platform for surveillance, communications and sensor payloads on sites with no mains supply and no fuel convoy.

Solar and battery carry the payload in silence. A Stage V generator running on HVO renewable diesel engages only to recharge the pack, guaranteeing uptime through low light, deep winter and heavy loads. The platform is delivered open: the universal head mount and internal equipment bay accept the operator's own cameras, lighting, sensors and radios.

KEY FEATURES

- 6 m hot-dip galvanised manual mast, mechanical safety brake; automatic patent winch brake available
- Hybrid energy pack: 24 VDC 220 Ah AGM battery with Stage V backup generator on HVO
- 3 × 360 W book-opening solar array, manual deployment, no tools
- Generator engages at 40% state of charge and shuts down once recharged
- Open head mount compatible with all major camera and lighting systems
- Lockable steel compartment for operator electronics, 370 × 200 × 470 mm
- Silent, zero-emission operation in battery mode
- Forklift pockets, all-terrain transport, ten units per truck



1,840 h	41 h	1,080 W	6 m
TOTAL RUNNING TIME BEFORE REFUELLING	BATTERY ONLY, GENERATOR IDLE	INSTALLED SOLAR ARRAY	MANUAL MAST HEIGHT

Autonomy figures based on continuous 24/7 operation at 100 W payload draw.

DEPLOYMENT

Remote construction & highways Compounds and corridors with no supply.	Oil, gas & mining Isolated assets, long service intervals.	Defence & border surveillance Low signature, full autonomy.
Critical national infrastructure Pipelines, substations, dams, telecoms.	Secured public & private zones Events, compounds, restricted areas.	Environmental monitoring Dust, gas, weather and traffic sensing.

MAST		RUNNING TIME AT 100 W	
Lifting method	Manual	Battery only	41 hrs
Maximum height	6 m	Total before refuelling	1,840 hrs
Rotation	None	DIMENSIONS & WEIGHT	
Safety brake	Mechanical	Transport L × W × H	1200 × 1700 × 2580 mm
Body	Hot-dip galvanised	In operation L × W × H	3280 × 2500 × 6000 mm
BATTERY		Weight	1,368 kg
Type & number	AGM × 2	Equipment bay, internal	370 × 200 × 470 mm
Power pack	24 VDC – 220 Ah	TRANSPORT & HANDLING	
Expected cycles	500	– Ten units per truck, packaged for rapid unloading	
Solar array	3 × 360 W, manual	– Forklift pockets; transportable across all terrains	
GENERATOR SET		– Mast raised manually in minutes by one operator, no tools or electric motors	
Engine	Kubota Z482 Stage V	– Solar array opens by hand; no technicians required on site	
Speed	1,500 rpm		
Alternator	Linz Electric 3.5 kVA		
Output	220 V – 50 Hz		
Fuel	HVO renewable diesel		

YOUR PAYLOAD, OUR POWER

Delivered as an open, ready-to-use platform. Operators integrate their own systems, or Melrakki specifies them:

- HD CCTV, PTZ and IP cameras
 - LTE, 5G and Wi-Fi router modules
 - Motion and environmental sensors
 - Remote access panel and software
- LED matrix or infrared floodlighting
 - GPS tracking and geofencing
 - Alarms, sirens and UPS
 - Operator branding and colour

COMMON QUESTIONS

- What makes the platform hybrid?**

Battery-first operation with solar recharge; the generator engages at 40% state of charge to recharge the pack, then stops.
- How loud is it?**

Silent on battery. The generator is low-RPM and noise-insulated.
- Does it need routine attendance?**

No. It is designed for long unattended operation on remote sites.
- Is it secure against tampering?**

All sensitive components are housed in a lockable steel compartment.



Tell us the load, the location and the conditions.

We engineer the platform that keeps it alive, and support it for the life of the deployment.

ENQUIRIES

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SUPPLY & SUPPORT

Lead time, commissioning, spares and warranty terms confirmed at quotation. Technical support direct from Melrakki Systems.